

CONDEP INJECTION 50mg/ml (3ml vial)**DESCRIPTION:**

CONDEP INJECTION 50mg/ml (3ml vial) is white to off-white sterile suspension for injection.

COMPOSITION:

Each ml contains Medroxyprogesterone Acetate 50mg.

PHARMACODYNAMICS:

Animal: Medroxyprogesterone acetate induces responses in laboratory animals comparable to those caused by progesterone. It is more potent than progesterone and, when injected as a suspension, has a long duration of action. Medroxyprogesterone acetate induces glandular development in the endometrium, maintains pregnancy, delays parturition, inhibits ovulation and suppresses oestrous cycles. Androgenic and anabolic effects have been noted, but the drug is apparently devoid of significant oestrogenic activity. In selected animal tests it has some adrenocorticoid-like activity and in dogs increases serum growth hormone levels. **Human:** Medroxyprogesterone acetate is a progestational agent with prolonged effect when administered by IM injection. When administered 3 monthly in recommended dose to women with adequate endogenous oestrogen, it transforms proliferative into secretory endometrium. Medroxyprogesterone acetate inhibits gonadotropin production, which in turn prevents follicular maturation and ovulation. These actions produce the contraceptive effect. The effectiveness is dependent on the women returning every month for injection. Women with lower body weights conceive sooner than women with higher body weights after discontinuation of Medroxyprogesterone acetate Injection.

Based on results from a French epidemiological case-control study, an association between medroxyprogesterone acetate and meningioma has been observed. This study was based on data from the French National health data system (SNDS – Système National des Données de Santé) and included a population of 18,061 women who had intracranial surgery for meningioma and 90,305 women without meningioma. The exposure to medroxyprogesterone acetate 150 mg/3ml injectable was compared between women who had intracranial surgery for meningioma and women without meningioma. Analyses showed an excess risk of meningioma with the use of medroxyprogesterone acetate 150 mg/3 ml (9/18,061 (0.05%) v 11/90,305 (0.01%), OR 5.55 (95%CI 2.27 to 13.56)). This excess risk seems to be driven primarily by prolonged use (≥3 years) of medroxyprogesterone acetate.

PHARMACOKINETICS:

Parenteral Medroxyprogesterone Acetate is a long acting progestational steroid because of its slow absorption from injection site.

Following a single 150mg intramuscularly dose of Medroxyprogesterone acetate levels increase for approximately 3 weeks to reach peak plasma concentration of 1 to 7 ng/mL. The level then decrease exponentially until they become undetectable (<100pg /ml) between 120 days and 200 days following injection. Considerable interindividual variability in serum levels occurs after administration of standard doses of intramuscular medroxyprogesterone acetate. Medroxyprogesterone acetate is metabolized and conjugated in the liver. Metabolic products are predominantly excreted in the urine, both as conjugated and free form.

INDICATIONS:

1. For contraception (ovulation suppression).
2. In the treatment of endometriosis – As adjunctive and/or palliative treatment of recurrent and/or metastatic endometrial or renal carcinoma.
3. In the treatment of hormonally-dependent, recurrent breast cancer in postmenopausal women.

RECOMMENDED DOSAGE:

Indications	Dosage
Inoperable, recurrent, metastatic, endometrial & renal carcinoma	Doses of 400MG to 1000MG of CONDEP intramuscularly per week are recommended initially. If improvement is noted within a few weeks or months and the disease appears stabilized, it may be possible to maintain improvement with as little as 400MG/month.
Endometriosis	50 mg weekly or 100mg every two weeks by intramuscular injection for at least six months.
Breast Carcinoma	IM injection 500mg daily for 4 weeks then 500MG twice weekly for maintenance.
Contraception (Ovulation suppress)	150 mg every 3 months administered by deep IM injection in the gluteal or deltoid muscle. To increase assurance that the patient is not pregnant at the time of the first administration, it is recommended that this injection is given only: – during the first 5 days after the onset of a normal menstrual period – within 5 days post-partum if not breastfeeding, or – if breastfeeding at 6-week post-partum, after having excluded pregnancy. If the period between injections is > 14 weeks, the physician should determine that the patients is not pregnant before re-administering the drug*.

Gluteal infiltration and abscess formation may occur with intramuscular administration. This complication appears to be particularly related to the volume administered and careful attention to injection technique should be observed. If large volume are to be given, i.e greater than 2.5 ml, the divided administration into several sites is recommended. It is also important that the suspension be shaken well before use and administered by deep intra-muscular injection into the gluteal muscle.

Routine or long-term cyclic use of supplemental oestrogens medroxyprogesterone acetate is not recommended. Excessive or prolonged bleeding which becomes troublesome to the patient can usually be controlled by the administration of oral or parenteral oestrogens in the equivalent of 0.05 to 0.1 mg ethinyl estradiol daily for 7 to 21 days. This therapy can be continued for 1 to 2 cycles, but should not be considered for long term administration.

If abnormal bleeding persists, appropriate investigation should be instituted to rule out the possibility of organic pathology.

CONTRAINDICATIONS:

1. In patients with a known sensitivity to medroxyprogesterone acetate or any ingredient of the vehicle.
2. During pregnancy, either for diagnosis or therapy.
3. As a contraceptive at the above dosage in known or suspected hormone-dependent malignancy of breast or genital organs.
4. Whether administered alone or in combination with oestrogen, CONDEP should not be employed in patients with abnormal uterine bleeding until a definite diagnosis has been established and the possibility of genital tract malignancy eliminated.
5. Patient with meningioma or history of meningioma (for non-oncological indications)

ROUTE OF ADMINISTRATION: Intramuscular (IM)

WARNING & PRECAUTIONS:**Warnings:**

1. **Thromboembolic Disorders:** The physician should be alert to the earliest manifestations of thrombotic disorders (thrombophlebitis, cerebrovascular disorders, pulmonary embolism, and retinal thrombosis). Should any of these occur or be suspected, the drug should be discontinued immediately.
2. **Ocular Disorders:** Discontinue medication pending examination if there is sudden partial or complete loss of vision, or if there is a sudden onset of proptosis, diplopia or migraine. If examination reveals papilloedema, or retinal vascular lesions, medication should be withdrawn.
3. **Bleeding Irregularities:** Most women receiving Medroxyprogesterone Acetate for contraception experienced disruption of menstrual bleeding patterns. Altered bleeding patterns including irregular or unpredictable bleeding or spotting, or rarely, heavy or continuous bleeding. If abnormal bleeding persists or is severe, appropriate investigations should be instituted to rule out the possibility of organic pathology and appropriate treatment should be instituted when necessary.
As women continued to use Medroxyprogesterone Acetate, fewer experienced intermenstrual bleeding and more experience amenorrhoea. By month 12, amenorrhoea was reported by 57% of women, and by month 24, amenorrhoea was reported by 68% of women using Medroxyprogesterone Acetate. Following repeated injections, amenorrhoea and infertility may persist for periods of up to 18 months and occasionally longer.
4. **Bone Mineral Changes:** It has been suggested that intramuscular medroxyprogesterone acetate may be considered one of the risk factors for the development of osteoporosis. The rate of bone loss is greatest in the early years of use and then subsequently approaches the normal rate of age-related fall.
5. **Cancer Risks:** Long-term case-controlled surveillance of Medroxyprogesterone Acetate use for contraception found no increased overall risk of breast cancer and no increased overall risk of ovarian, liver, or cervical cancer.
6. **Accidental Pregnancies:** Infants from accidental pregnancies that occur 1-2 months after injection of Medroxyprogesterone Acetate may be at increased risk of low birth weight, which in turn may be associated with an increased risk of neonatal death. The attributable risk is low because such pregnancies are uncommon. A significant increase in polysyndactyly and chromosomal anomalies was observed among infants of Medroxyprogesterone Acetate users, the former being most pronounced in women under 30 years of age. The unrelated nature of these defects, the lack of confirmation from other studies, the distant preconceptual exposure to Medroxyprogesterone Acetate, and the chance effects due to multiple statistical comparisons, make a causal association unlikely.
7. **Ectopic Pregnancy:** As with all forms of hormonal contraceptions, health-care providers should be alert to the possibility of an ectopic pregnancy among women using Medroxyprogesterone Acetate who become pregnant or complain of severe abdominal pain.
8. **Sexually Transmitted Diseases:** Medroxyprogesterone Acetate 150 mg/mL is intended to prevent pregnancy. It does not protect against HIV infection (AIDS) and other sexually transmitted diseases. The woman should be advised that additional measures are needed to prevent the transmission of sexually transmitted diseases.
9. **Meningioma:** Cases of meningioma (single and multiple) have been reported in patients treated with medroxyprogesterone acetate for a prolonged time (several years). Patients treated with medroxyprogesterone acetate should be monitored for signs and symptoms of meningioma in accordance with clinical practice.
In some cases, shrinkage of meningioma was observed after treatment discontinuation of depot medroxyprogesterone acetate. If a patient treated for a non-oncological indication is diagnosed with meningioma, medroxyprogesterone acetate must be stopped, as a precautionary measure.
If a patient treated for an oncological indication is diagnosed with meningioma, the need for further treatment with medroxyprogesterone acetate should be carefully considered on a case-by-case basis taking into account individual benefits and risks.
10. In all situations where cessation of therapy is warranted, the physician should be aware of the slow elimination of the depot formulation.
11. Clinical suppression of adrenocorticoid function has not been observed at low dose levels, however at the high doses used in the treatment of cancer, corticoid-like activity has been reported. Animal studies show that medroxyprogesterone acetate possesses adrenocorticoid activity.
12. Anaphylactic and anaphylactoid reactions have occasionally been reported in patients treated with intramuscular medroxyprogesterone acetate.
13. The following laboratory tests may be affected by the use of Medroxyprogesterone Acetate:
 - a) Gonadotrophin levels
 - b) Plasma progesterone levels
 - c) Urinary pregnanediol levels
 - d) Plasma testosterone levels (in the male)
 - e) Plasma oestrogen levels (in the female)
 - f) Plasma cortisol levels
 - g) Glucose Tolerance Test
 - h) Metyrapone Test.
 - i) Sex-hormone-binding-globulin concentrations are decreased.
 - j) Coagulation test values for prothrombin (Factor II) and Factors VII, VIII, IX and X may increase.

Precautions:

1. **Physical Examination:** The pretreatment physical examination should include special reference to breast and pelvic organs, as well as Papanicolaou smear.
2. **Fluid Retention:** Because this drug may cause some degree of fluid retention, conditions which might be influenced by this factor, such as epilepsy, migraine, asthma, or cardiac or renal dysfunction, require careful observation.
3. **Breakthrough Bleeding:** Breakthrough bleeding is likely to occur in patients being treated for endometriosis. No other hormonal intervention is recommended for managing this bleeding. Non-functional causes should also be borne in mind and in cases of undiagnosed vaginal bleeding, adequate diagnostic measures are indicated.
4. **Carbohydrate Metabolism:** A decrease in glucose tolerance has been observed in some patients on progestogens. The mechanism of this decrease is obscure. For this reason, diabetic patients should be carefully observed while receiving progestogen therapy.
5. **CNS Disorders and Convulsions:** Patients who have a history of mental depression should be carefully observed and the drug discontinued if the depression recurs to a serious degree.
6. **Weight Changes:** There is a tendency for women to gain weight while on Medroxyprogesterone Acetate therapy. From an initial average body weight of 61.8 kgs (136 lbs) women who completed 1 year of therapy with Medroxyprogesterone Acetate gained an average of 2.45 kgs (5.4 lbs). Women who completed 2 years of therapy gained an average of 3.68 kgs (8.1 lbs).
Women who completed 4 years gained an average of 6.3 kgs (13.8 lbs). Women who completed 6 years gained an average of 7.5 kgs (16.5 lbs).
Two per cent of women withdrew from a large-scale clinical trial because of excessive weight gain.
7. **Return of Fertility:** Medroxyprogesterone Acetate has a prolonged contraceptive effect. There is no evidence that CONDEP causes permanent infertility. Pregnancies have occurred as early as 14 weeks after a preceding injection.
8. **Liver Function:** Certain endocrine and possible liver function tests may be affected by treatment with Medroxyprogesterone Acetate. Therefore, if such tests are abnormal in a patient taking Medroxyprogesterone Acetate, it is recommended that they be repeated after the drug has been withdrawn.
9. **Patient Age:** The age of the patient constitutes no absolute limiting factor although treatment with progestins may mask the onset of the climacteric.
10. **Pathology Tests:** The pathologist should be advised of progestin therapy when relevant specimens are submitted.
11. **IM Administration:** Gluteal infiltration and abscess formation may occur with intramuscular administration. See Dosage and Administration.

12. Because of the prolonged action and the resulting difficulty in predicting the time of withdrawal bleeding following injection, Medroxyprogesterone Acetate is not recommended for treatment for secondary amenorrhoea or dysfunctional uterine bleeding. In these conditions, oral therapy is recommended.

INTERACTION WITH OTHER MEDICATIONS:

Aminoglutethimide administered concomitantly with Medroxyprogesterone Acetate may significantly decrease the serum concentration of medroxyprogesterone acetate. Medroxyprogesterone Acetate users should be warned of the possibility of decreased efficacy with the use of this or any related drugs.

PREGNANCY & LACTATION:

Use in pregnancy: Studies in animals have shown that progestogens, including medroxyprogesterone acetate, may have an adverse effect on the developing fetus, including teratogenicity and fetotoxicity. In addition, other animal studies have shown that high doses of progestogens can cause masculinisation of the female fetus. Several reports suggest an association between intrauterine exposure to progestational drugs in the first trimester of pregnancy and genital abnormalities in male and female fetuses. The risk of hypospadias (5 to 8 per 1000 male births in the general population) may be approximately doubled with exposure to these drugs. There are insufficient data to quantify the risks to female fetuses, but because some of these drugs induce mild virilisation of the external genitalia of the female fetus and because of the increased association of hypospadias in the male fetus, it is prudent to avoid use of these drugs during the first trimester of pregnancy.

Children exposed to medroxyprogesterone acetate in utero and followed to adolescence, showed no evidence of any adverse effects on their health including their physical, intellectual, sexual or social development.

Medroxyprogesterone Acetate IS NOT TO BE USED AS A TEST FOR PREGNANCY OR WHERE PREGNANCY IS SUSPECTED.

To ensure that Medroxyprogesterone Acetate is not administered inadvertently to a pregnant woman, it is important that the first injection only be given:

- during the first five days after the onset of a normal menstrual period
- within five days post-partum if not breast feeding and
- if breast feeding, at the sixth week post-partum, after having excluded pregnancy

Use in lactation: Detectable amounts of the drug appear in breast milk. Milk composition, quality and amount are not adversely affected in nursing mothers treated with Medroxyprogesterone Acetate injection. No adverse development or behavioral effects have been noted in infants exposed to Medroxyprogesterone Acetate via breast milk and studied through puberty.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES: The effect of medroxyprogesterone acetate on the ability to drive and use machinery has not been systematically evaluated.

SIDE EFFECTS:

The following events listed in order of seriousness rather than frequency of occurrence, have been associated with the use of progestogens including Medroxyprogesterone.

1. Anaphylaxis and anaphylactoid-like reactions.
2. Thromboembolic disease-Thrombophlebitis and pulmonary embolism.
3. Central Nervous System-Nervousness, insomnia, somnolence, fatigue, depression, dizziness and headache and tremor.
4. Gluteal infiltration and abscess formation. This reaction appears to be related to the volume of agent administered at the highest frequency of this complication occurs with large volumes, i.e greater than 2.5 ml.
5. Skin and mucous membranes-Urticaria, pruritus, rash, acne hirsutism, alopecia and sweating.
6. Gastro-Intestinal nausea.
7. Breast-Tenderness and galactorrhoea.
8. Cervix-Changes in position of the transformation zone and secretions.
9. Miscellaneous-Hyperpyrexia, Cushing Syndrome, weight gain.
10. Moderate elevation of blood pressure, transient elevation of alkaline phosphatase and/or serum transaminase activities, elevations of serum calcium and potassium levels, and increased in white cells and platelet counts.
11. Neoplasms benign, malignant & unspecified: Frequency (Unknown) - meningioma

SYMPTOMS AND TREATMENT OF OVERDOSE:

Patients receiving pharmacological doses of medroxyprogesterone acetate for treatments of neoplasms (400 mg/day or greater) may occasionally exhibit effects resembling those of glucocorticoid excess. No serious medical effects have been reported in association with over-dosage of medroxyprogesterone acetate injection.

INCOMPATIBILITIES: Not applicable

STORAGE CONDITIONS: Store below 30°C. Protect from light and freezing. Shake well before use. Keep out of reach of children. JAUHKAN DARIPADA KANAK-KANAK.

PACK SIZE: Box of 10's vials x 3ml

SHELF LIFE: Please refer outer package.

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599 Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang,
Selangor Darul Ehsan, MALAYSIA.